

# PAPER\_1329\_GALAXY\_BAR\_FRACTION

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## PAPER\_1329 — Galaxy Bar Fraction $f_{\text{bar}}$

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**Framework:** UQFF — Star-Magic v5.27+

**Tier:** D — Astrophysics / Cosmology Open (CLOSED)

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**Calculator:** `calculate_paradox({"paradox": "galaxy_bars"})`

**Master Expression** ``  $f_{\text{bar}} = \Phi_{\text{res}} \times \beta_i = 50.6\%$  `` **\*\*Match: within obs 30-50% band\*\***

**Interpretation** Bar formation in disk galaxies driven by  $\Phi_{\text{res}} \times \beta_i$  resonant coupling.

**UQFF Locked Primitives** -  $D_{\text{phys}} = 4$ ,  $D_{\text{BSFG}} = 6$ ,  $D_{\text{crit}} = 26$ ,  $SO_5 = 10$ ,  $A_5 = 60$ ,  $N_{\text{CH}} = 9$  -  $K_{\text{MEX}} = 25/12$ ,  $\beta_i = 0.6029$ ,  $\Phi_{\text{res}} = 0.84$ ,  $\Lambda = 0.00729735$ ,  $F_{\text{TRZ}} = 0.1$

**NOT REPLACEMENT UQFF** derives via integer-primitive identity. Zero fit parameters.

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